

A Mini Project Report
on
PLASTIC DETECTION USING YOLO
BACHELOR OF ENGINEERING
in
ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

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2025-2026

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DECLARATION BY THE CANDIDATES

We, **Bharath Simha Reddy V (160723748001)**, **Raavi Indu (160723748030)** and **Korimi Akash (160723748031)** students of Methodist College of Engineering and Technology, pursuing Bachelor's degree in Department of Artificial Intelligence and Machine Learning, hereby declare that this Mini Project report entitled "**PLASTIC DETECTION USING YOLO**", carried out under the guidance of **Mr. Mujthaba Gulam Muqeeth** submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Department of Artificial Intelligence and Machine Learning. This is a record work carried out by us and the results embodied in this report have not been reproduced/copied from any source.

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CERTIFICATE BY THE SUPERVISOR

This is to certify that this Mini Project work entitled “**PLASTIC DETECTION USING YOLO**” by **Bharath Simha Reddy V (160723748001)**, **Raavi Indu (160723748030)** and **Korimi Akash (160723748031)** submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Department of Artificial Intelligence and Machine Learning, during the academic year 2025-2026, is a Bonafide record of work carried out by them.

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CERTIFICATE BY HEAD OF THE DEPARTMENT

This is to certify that this Mini Project work entitled “**PLASTIC DETECTION USING YOLO**” by **Bharath Simha Reddy V (160723748001)**, **Raavi Indu (160723748030)** and **Korimi Akash (160723748031)** submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Department of Artificial Intelligence and Machine Learning, during the academic year 2025-2026, is a Bonafide record of work carried out by them.

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PROJECT APPROVAL CERTIFICATE

This is to certify that this Mini Project work entitled **“PLASTIC DETECTION USING YOLO”** by **Bharath Simha Reddy V (160723748001)**, **Raavi Indu (160723748030)** and **Korimi Akash (160723748031)** submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Department of Artificial Intelligence and Machine Learning during the academic year 2025-2026, is a bonafide record of work carried out by them.

INTERNAL

EXTERNAL

HOD

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Vision & Mission

VISION

To become a leader in providing Computer Science & Engineering education with emphasis on knowledge and innovation.

MISSION

- M1:** To offer flexible programs of study with collaborations to suit industry needs
- M2:** To provide quality education and training through novel pedagogical practices
- M3:** To Expedite high performance of excellence in teaching, research and innovations.
- M4:** To impart moral, ethical valued education with social responsibility.

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Graduates of Compute Science and Engineering at Methodist College of Engineering and Technology will be able to:

- PEO1:** Apply technical concepts, Analyze, synthesize data to Design and create novel products and solutions for the real-life problems.
- PEO2:** Apply the knowledge of Computer Science Engineering to pursue higher education with due consideration to environment and society.
- PEO3:** Promote collaborative learning and spirit of team work through multidisciplinary projects
- PEO4:** Engage in life-long learning and develop entrepreneurial skills.

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- PSO2:** Develop software applications with open-ended programming environments.
- PSO3:** Design and develop solutions by following standard software engineering principles and implement by using suitable programming languages and platforms

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At the end of the four-year Bachelor of Engineering Program, engineering graduates will be able to:

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PO2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

ABSTRACT

Plastic pollution has emerged as one of the most critical environmental challenges, significantly impacting marine ecosystems, biodiversity, and long term ecological balance. One of the major difficulties in addressing this issue is the lack of efficient automated systems capable of accurately detecting and localizing plastic waste in real-world environments, particularly in underwater and marine settings where visibility and object complexity are high.

To address this problem, we propose an intelligent system titled “Plastic Detection using YOLO”, which leverages deep learning-based object detection techniques to identify and localize marine plastic debris in images. Unlike traditional classification approaches, the system uses a YOLO (You Only Look Once) model to perform real-time object detection, generating bounding boxes around detected debris along with confidence scores indicating prediction reliability.

The model is trained on the TrashCan dataset, which contains annotated underwater images of marine waste. Given an input image through a Streamlit based web interface, the system processes it using the trained YOLO network and outputs the locations of plastic debris within the scene. Each detection is visually represented using bounding boxes, enabling precise spatial understanding of waste distribution.

This approach enhances automation in environmental monitoring by enabling both detection and localization of plastic waste, making it more suitable for real-world deployment scenarios such as autonomous ocean-cleaning systems and robotic inspection platforms. Ultimately, the proposed system contributes toward scalable marine pollution monitoring and supports technological efforts aimed at preserving aquatic ecosystems.

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